

End-of-studies Apprenticeship Report

Optimizing and Adapting Language Models for Domain-Specific Task

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Presentation Outline

Problem 1: Integration of External Knowledge for Language Models

Problem 2: Behavior Adaptation for Small Models

Conclusion

From research to practice: challenges of AI implementation in business

Reality Check: 95% of AI generation/LLM projects in companies fail to generate significant P&L value.

Two Fundamental Challenges in Domain-Specific NLP:

Knowledge Gap

- ▶ Static training data
- ▶ Hallucination risks
- ▶ No access to proprietary documents

Behavior Gap

- ▶ Cannot follow domain workflows
- ▶ Out-of-character drift

How can we bridge these gaps with resource-constrained deployment?

Bridging the Gaps: Two Complementary Approaches

Addressing Knowledge Gap

Common Approaches:

- ▶ Continued pre-training
- ▶ Prompt engineering
- ▶ **Retrieval-Augmented Generation (RAG)**
- ▶ API integration

Addressing Behavior Gap

Common Approaches:

- ▶ Complex prompt engineering
- ▶ Few-shot learning
- ▶ **Fine-tuning**
- ▶ RLHF

This work explores both approaches with real-world applications

Problem 1

Integration of External Knowledge for Language Models

Building a RAG System for Complex Documents

Problem 1: Integration of External Knowledge for Language Models

Use Case: Building a domain-specific QA system for financial and healthcare documents

- ▶ **Objective:** Enable accurate question answering over complex, proprietary documents
- ▶ **Constraints:** High accuracy, low latency, cost-effective deployment

Key Questions:

- ▶ How to preserve structure in complex documents?
- ▶ How to ensure high accuracy with limited resources?
- ▶ How to handle multimodal data effectively?

The Real Challenge: Complex Financial Data

Case Study 1: Financial - GPM Reports

Complex financial tables and charts

Le **chiffre d'affaires** de la ligne d'activité « santé assimilée à la vie » s'élève à 9 752 k€, en hausse de 2% à la fin 2024 par rapport à la fin 2023.

La **charge de sinistres** de la ligne d'activité « santé assimilée à la vie » s'établit à 6 410 k€ en 2024, soit une hausse de 25% par rapport à l'exercice 2023.

La **cession** a un solde négatif de 3 540 k€ en 2024, en hausse de 7% par rapport à l'exercice 2023 et expliquée par la mise en place en 2022 d'un traité de réassurance avec Allianz.

Les **frais** s'élèvent 3 084 k€ en 2024, en baisse de 14% par rapport à la fin 2023.

En conséquence, le **résultat technique net** de frais et de réassurance de la ligne d'activité « santé assimilée à la vie » s'établit à - 3 282 k€ au titre de l'exercice 2024 contre - 2 504 k€ en 2023, soit une baisse de 31%.

A.2.2.2. Résultat Rentes

Résultat de souscription (en k€)	31/12/2024	31/12/2023	Variation en %
Primes brutes	43 899	40 476	8%
Charges sinistres brutes	15 609	13 594	15%
Résultat technique brut	28 290	26 882	5%
Primes cédées	12 630	11 728	8%
Charges sinistres cédées	3 523	2 636	34%
Résultat technique cédé	9 107	9 082	0%
Primes nettes	31 269	28 747	9%
Charges sinistres nettes	12 086	10 958	10%
Résultat technique net	19 183	17 789	8%
Frais administratifs	6 451	7 883	-18%
Frais de gestion des investissements	115	118	-3%
Frais de gestion des sinistres	991	1 005	-1%
Frais d'acquisition	4 170	4 566	-9%
Frais généraux	-	-	n.s

Objective

- ▶ Quick lookup of information from annual financial reports

Challenges

- ▶ Multi-modal data: balance sheets, tables, charts, images
- ▶ Long documents (50-200 pages)
- ▶ Preserving relationships between data points

The Real Challenge: Complex Healthcare Data

Case Study 2: Healthcare - CCAM Catalog

Thousands of medical codes requiring exact matching

Objective

- Find the correct medical code(s) for given clinical scenarios

Challenges

- Thousands of codes in complex tabular structure
- Exact code matching required
- Similar code descriptions causing confusion

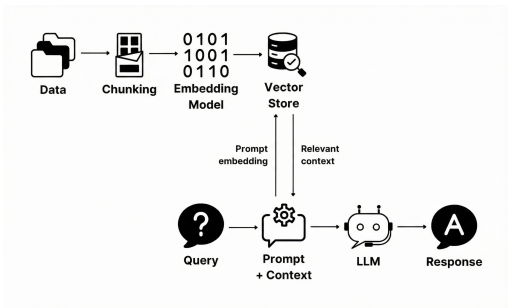
CCAM - CCAM version 100 applicable 01/01/2021									
Code	Texte	Activité	Phase	Tarif Secteur 1 / adhérent OPTAMOPTAM-ACD (en euro)	Tarif Hors secteur 1 / OPTAMOPTAM-ACD (en euro)	Remb. par Couv.	Accréd. Française	En V	Dégré équival.
1	SYSTÈME NERVEUX CENTRAL, PÉRIPHÉRIQUE ET AUTONOME								
	À l'exclusion de : - actes de diagnostic - actes de traitement - actes de rééducation - actes de soins de suite et de réadaptation - actes de soins de palliatifs - actes de soins de confort - actes de soins de confort et de rééducation - actes de soins de confort et de rééducation et de soins de suite et de réadaptation - actes de soins de confort et de rééducation et de soins de suite et de réadaptation et de soins de palliatifs - actes de soins de confort et de rééducation et de soins de suite et de réadaptation et de soins de palliatifs et de soins de confort								
1.01	ACTES DIAGNOSTIQUES SUR LE SYSTÈME NERVEUX								
	À l'exclusion de : - actes de diagnostic - actes de traitement - actes de rééducation - actes de soins de suite et de réadaptation - actes de soins de palliatifs - actes de soins de confort - actes de soins de confort et de rééducation - actes de soins de confort et de rééducation et de soins de suite et de réadaptation - actes de soins de confort et de rééducation et de soins de suite et de réadaptation et de soins de palliatifs - actes de soins de confort et de rééducation et de soins de suite et de réadaptation et de soins de palliatifs et de soins de confort								
01.01.01	Explorations électrophysiologiques du système nerveux								
01.01.01.01	Electromyographie (EMG)								
	Facturation : les examens électrophysiologiques doivent être facturés avec un appareil comportant un système d'enregistrement permettant en, diffus, une durée qualitative et quantitative								
AUQ001	Electromyographie par électrode de surface, sans enregistrement vidéo	1	0	Non pris en charge	Non pris en charge				1 ATU
AUQ002	Electromyographie par électrode de surface, avec enregistrement vidéo	1	0	Non pris en charge	Non pris en charge				1 ATU
AUQ003	Electromyographie de 1 ou 2 muscles actifs au repos et à l'effort sans stimulation, par électrode aiguille	1	0	51,84	51,84				1 ATU
AUQ004	Electromyographie de 1 ou 2 muscles actifs au repos et à l'effort avec stimulation, par électrode aiguille	1	0	86,40	86,40				1 ATU
[P, P, S, U]	Electromyographie de 3 à 5 muscles actifs au repos et à l'effort sans stimulation, par électrode aiguille	1	0	65,83	65,83				1 ATU
AUQ005	Electromyographie de 3 à 5 muscles actifs au repos et à l'effort avec stimulation, par électrode aiguille	1	0	86,40	86,40				1 ATU
[P, P, S, U]	Electromyographie de 6 à 8 muscles actifs au repos et à l'effort sans stimulation, par électrode aiguille	1	0	74,72	74,72				1 ATU
AUQ006	Electromyographie de 6 à 8 muscles actifs au repos et à l'effort avec stimulation, par électrode aiguille	1	0	124,82	124,82				1 ATU

Other Technical Barriers

- ▶ **Strict data security constraints:** Sensitive and private business data can not be exposed to ChatGPT or Cloud Services
- ▶ **High accuracy requirements:** Hallucination is unacceptable in high-stakes domains like finance and healthcare
- ▶ **Resource-constrained deployment:** Large models require significant computational resources, while smaller models have limited context windows

→ **Solution Objective:** A unified in-house RAG pipeline that can handle complex documents with high accuracy

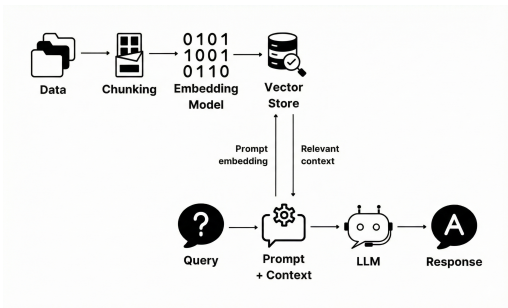
Simple RAG Pipeline - Baseline Approach



Simple RAG Configuration

- ▶ **Parsing:** PyPDF for text extraction
- ▶ **Chunking:** Fixed-size (512 tokens/chunk, 50 overlap)
- ▶ **Embedding:** all-MiniLM-L6-v2
- ▶ **Vector Store:** ChromaDB with Dense Search
- ▶ **Generation:** Gemma-3-12b-it

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Why Simple RAG Fails: Issue 1 - Loss of Table Structure

Problem: Flattening of 2D Table Structure

Original Table:

Sous modules (en k€)	SCR 2024	SCR 2023	Évol.
Type 1	2 692	2 487	8 %
Type 2	28 377	36 221	-22 %
Diversification	-621	-586	6 %
Risque de défaut	30 448	38 121	-20 %



PyPDF Output:

Sous modules (en k€) SCR 2024 SCR 2023 Evolution
Type 1 2 692 2 487 8 %
Type 2 28 377 36 221 -22 %
Effet de diversification -621 -586 6 %
Risque de défaut 30 448 38 121 -20 %

Impact: Row/column context is lost, making it difficult for LLMs to understand relationships

Why Simple RAG Fails: Issue 2 - Failure with Exact Code Matching

Problem: Dense embedding struggles with exact keyword matching

User Query:

HAFA006



Search Results:

=====

ANALYSIS OF RESULTS

=====

1 HAFA006 POSITION:

✓ Found HAFA006 at rank #3

⚠ HYPOTHESIS PARTIALLY CONFIRMED: Ranked #3 (not #1, but still high)

2 SIMILAR CODES RANKED HIGHER:

Found 5 chunks with similar HAFA00X codes:

- Rank #1: ['HAFA004', 'HAFA002'] (score: 0.2544)
- Rank #2: ['HAFA005', 'HAFA003'] (score: 0.2315)
- Rank #3: ['HAFA007'] (score: 0.2199)
- Rank #4: ['HAFA001'] (score: 0.2107)
- Rank #5: ['HAFA009'] (score: 0.2107)

Impact: Exact code "HAFA006" not found in highest top result; semantically similar but incorrect codes ranked higher

Why Simple RAG Fails: Quantitative Analysis

Other Issues:

- ▶ **Lost in the middle:** LLM accuracy drops when ground truth passage is not in high position in top-k results
- ▶ **Multilingual embedding mismatch:** English embedding model applied to French documents

Quantitative Results on GPM Test Set

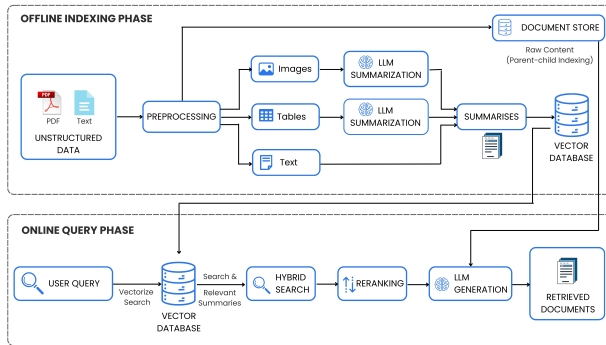
Metric	Score	Interpretation
Correctness	0.66	Only 66% answers correct w.r.t. contexts
Relevancy	0.73	73% answers relevant to the question

Conclusion: Simple RAG achieves only 66% correctness — insufficient for high-stakes domains

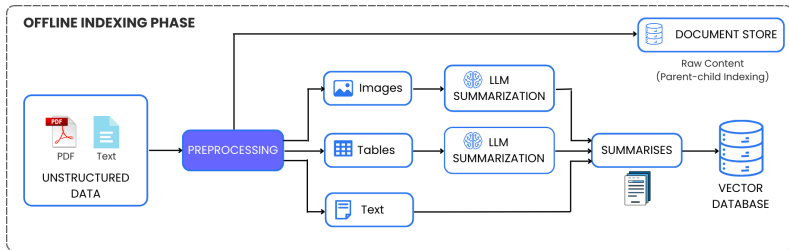
Advanced Multimodal RAG Pipeline - Overview

Key Innovations:

- ▶ Structure-preserving document processing
- ▶ Multimodal summarization
- ▶ Hybrid search combining semantic and keyword matching
- ▶ Cross-encoder reranking



Offline Indexing: Structure Preservation



Preserving Structure with UnstructuredIO and Table Transformer:

1. **Table Detection:** Identify table positions using YoloX/Detectron2
2. **Structure Recognition:** Recognize cells, merged cells, headers vs. body
3. **HTML Mapping:** Convert to `<table>` tags for easy LLM processing

Offline Indexing: Example of Structure Preservation

Le « type 2 » prend en compte dans son calcul du risque le montant des créances en distinguant les créances de plus et celles de moins de 3 mois. Il intègre les éléments suivants :

- les créances nées d'opérations diverses ;
- les créances d'impôts et taxes.

Les tableaux ci-après présentent la répartition des expositions et la répartition du SCR de contrepartie par typologie de contrepartie au 31 décembre 2024 :

Sous modules (en k€)	SCR 2024	SCR 2023	Evolution
Type 1	2 692	2 487	8 %
Type 2	28 377	36 221	-22 %
Effet de diversification	-621	-586	6 %
Risque de défaut	30 448	38 121	-20 %



Le risque de type 2 représente 91 % du SCR de contrepartie.

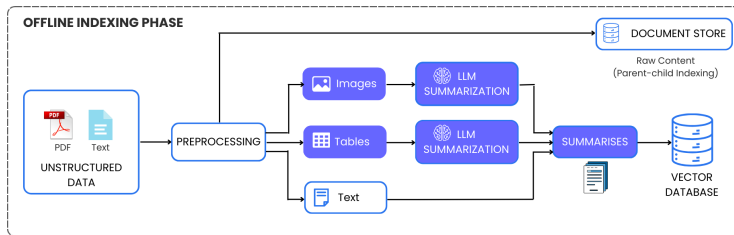
Le SCR de défaut baisse de 20% à la fin 2024 et s'explique par la baisse en 2024 des contreparties de type 2 (-22%).

Table extraction preserves spatial relationships

```
<table>
  <thead>
    <tr>
      <th>Sous modules (en k€)</th>
      <th>SCR 2024</th>
      <th>SCR 2023</th>
      <th>Evolution</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Type 1</td>
      <td>2 692</td>
      <td>2 487</td>
      <td>8%</td>
    </tr>
    <tr>
      <td>Type 2</td>
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    <tr>
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    </tr>
    <tr>
      <td>Risque de défaut</td>
      <td>30 448</td>
      <td>38 121</td>
      <td>-20%</td>
    </tr>
  </tbody>
</table>
```

HTML output with structure intact

Offline Indexing: Multimodal Summarization



Purpose: Reduce context size while preserving key information

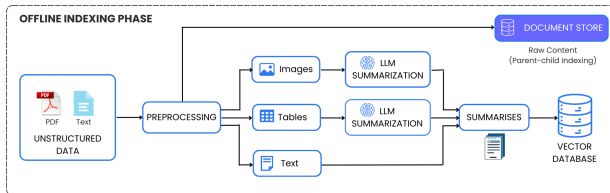
Original Table:

Sous modules	2024	2023	Évol.
Type 1	2 692	2 487	8 %
Type 2	28 377	36 221	-22 %
Diversif.	-621	-586	6 %
Total	30 448	38 121	-20 %



Summary: "Le tableau détaille l'évolution du Solvency Capital Requirement (SCR) pour différents sous-modules entre 2023 et 2024, avec les montants exprimés en milliers d'euros (k€) et l'évolution en pourcentage. Pour le sous-module « Type 1 », le SCR est passé de 2 487 k€ en 2023 à 2 692 k€ en 2024, affichant une évolution de 8%. Le sous-module « Type 2 » a vu son SCR passer de 36 221 k€ en 2023 à 28 377 k€ en 2024, avec une évolution indiquée de -22%. L'« Effet de diversification » a évolué de -586 k€ en 2023 à 621 k€ en 2024, soit une évolution de 6%. Enfin, le « Risque de défaut » a diminué, passant de 38 121 k€ en 2023 à 30 448 k€ en 2024, l'évolution diminuée de -20%."

Offline Indexing: Parent-Child Architecture



Design Principle:

- ▶ **Child (Summary):** Stored in vector store for efficient retrieval
- ▶ **Parent (Raw Content):** Stored in document store for detailed generation

Benefit: Retrieve using concise summaries, generate using complete information

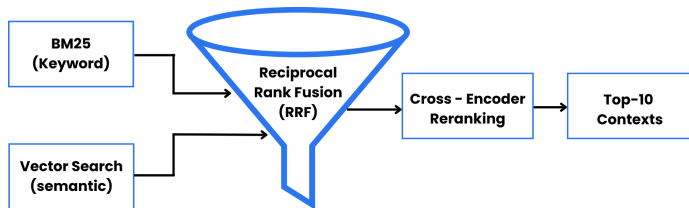
Offline Indexing: Multilingual Embedding Model

Rank (Bo...	Model	Zero-shot	Memory U...	Number of P...	Embedding D...	Max Tokens	Mean (T...	Mean (TaskT...	Bitext ...	Classification	Clustering
2	multilingual-e5-large-instruct	99%	1068	0.560	1024	514	63.22	55.08	80.13	64.94	50.75
3	embeddinggemma-300m	99%	1155	0.308	768	2048	61.15	54.31	64.40	60.90	51.17
4	bilingual-embedding-large	98%	2136	0.560	1024	514	60.96	52.92	73.55	62.77	46.49
5	Kalm-embedding-multilingual-mini-v1	92%	1885	0.494	896	512	57.05	50.05	64.77	57.57	45.61
6	Solon-embeddings-large-0.1	⚠ NA	2136	0.560	1024	514	59.63	52.01	76.10	60.84	43.86
7	bge-m3	98%	2167	0.568	1024	8194	59.56	52.18	79.11	60.35	40.88
8	gte-multilingual-base	99%	582	0.305	768	8192	58.34	51.50	71.79	57.23	44.23
9	jina-embeddings-v3	99%	1092	0.572	1024	8194	58.37	50.66	65.25	58.77	45.65
10	multilingual-e5-large	99%	2136	0.560	1024	514	58.62	51.42	73.81	59.43	42.69
11	Kalm-embedding-multilingual-mini-instruct-	92%	1885	0.494	896	512	56.35	49.28	64.22	57.38	45.89

Choice: Multilingual-E5-large-instruct

- ▶ 560M parameters, trained on 100+ languages
- ▶ Outperforms other models <1B params on MTEB multilingual benchmark
- ▶ Instruction-following capability for better query understanding

Online Querying Phase



Combining Two Approaches:

- ▶ **BM25 (Keyword):** Exact term matching for codes, names
- ▶ **Vector Search (Semantic):** Contextual understanding
- ▶ **Reciprocal Rank Fusion:** Merge results intelligently

Reranking Process:

1. Hybrid search retrieves candidate documents
2. Cross-encoder computes relevance scores for each query-document pair
3. Top-k most relevant documents sent to LLM for generation

Benefit: Addresses "lost in the middle" problem by ensuring most relevant context is highly ranked

Experimental Evaluation Setup

Test Datasets

- ▶ **GPM Test Set:** 25 complex questions on GPM's financial reports
- ▶ **CCAM Test Set:** 10 clinical consultation scenarios (provided by Dr. Besnier)

LLM-as-a-Judge Evaluation using RAGAS Metrics

- ▶ Two judge models: gemma-3-12b-it, mistral-small-24b-instruct-2501
- ▶ All metrics on continuous scale [0, 1], averaged across both judges

Metric	Definition
Answer Correctness	Semantic similarity: generated vs. ground-truth
Answer Relevancy	How well response addresses query
Faithfulness	Response grounded in retrieved context
Context Precision	Relevant chunks ranked highly
Code Hit Rate@k	Correct code in top-k results (CCAM only)

Results: Superior Accuracy Improvement

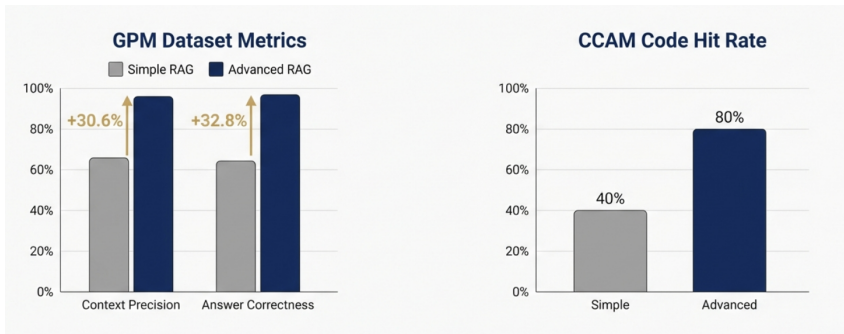
Table: Results on GPM Test Set (25 questions)

Pipeline	Correct.	Relev.	Faith.	Prec.
Simple RAG	0.648	0.788	0.732	0.654
Advanced RAG	0.976	0.972	0.940	0.960
<i>Improvement</i>	<i>+32.8%</i>	<i>+18.4%</i>	<i>+20.8%</i>	<i>+30.6%</i>

Table: Results on CCAM Test Set (10 scenarios)

Pipeline	Hit Rate@5	Hit Rate@20
Simple RAG	0.4	0.6
Advanced RAG	0.8	0.9
<i>Improvement</i>	<i>+40%</i>	<i>+30%</i>

Results: Visual Comparison



Key Observations:

- ▶ Advanced RAG achieves near-perfect performance on GPM dataset
- ▶ Biggest improvements in correctness and context precision
- ▶ Dramatic improvement in exact code retrieval for CCAM
- ▶ Consistent gains across all evaluation metrics

Ablation Study: Contribution of Each Component

Table: Incremental improvements on GPM Test Set

Configuration	Precision	Δ vs Base	Cumulative Δ
Baseline (Simple RAG)	0.648	—	—
+ UnstructuredIO Parsing	0.762	+11.4%	+11.4%
+ Summarization	0.818	+5.6%	+17.0%
+ Multilingual E5 Embedding	0.875	+5.7%	+22.7%
+ Hybrid Search	0.920	+4.5%	+27.2%
+ Reranking	0.976	+5.6%	+32.8%

Key Insights:

- ▶ Structure preservation (UnstructuredIO) yields the largest single gain (+11.4%)
- ▶ Each component contributes meaningfully to the final performance
- ▶ Cumulative effect demonstrates value of holistic pipeline design

Problem 2

Behavior Adaptation and Resource Optimization for Small Models

Fine-tuning Small Models for Specialized Tasks

Problem 2: Behavior Adaptation for Small Models

Context: Building a chatbot for Tarot Reading (psychological counseling)

Naive Approach: Prompt engineering with large LLMs (GPT-4, Claude-3) via APIs

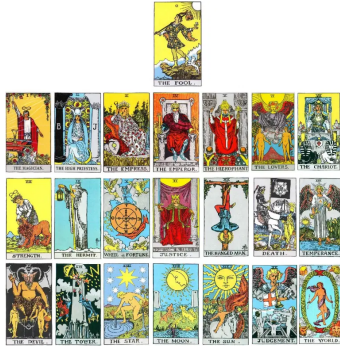
Why Prompt Engineering Alone Is Insufficient:

- ▶ **Attention dilution:** Important context lost in long prompts
- ▶ **Inconsistency:** Out-of-character drift in extended conversations
- ▶ **Cost:** Large models expensive for continuous deployment

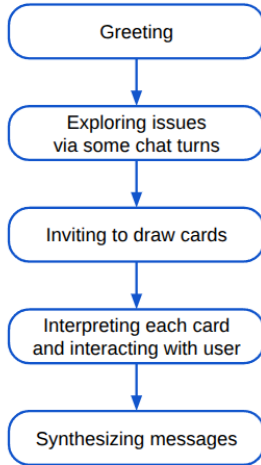
Table: Operational cost comparison. *10K conversations × 3K tokens each. **Amortized GPU cost.

Option	Cost/1K tokens	Cost/month*
GPT-4-turbo API	\$0.01–0.03	\$300–900
Claude 3 Sonnet API	\$0.003–0.015	\$90–450
Qwen-1.5B Fine-tuned	\$0.0005**	\$10–50
Qwen-0.5B Fine-tuned (Edge)	\$0	\$0

Specific Requirements for a Tarot Reader Task



Knowledge: Card meanings and spreads



Workflow: Strict conversation structure

Behavioral Traits:

- ▶ Empathetic tone
- ▶ Open-ended interaction
- ▶ Avoid direct predictions

Example of a Tarot conversation

Illustrative Example: Tarot Conversation

Querent: Hi, I've been feeling stuck in my job lately. I'm not sure if I should stay or look for something new.

Chatbot: Hey there! Career crossroads can be tricky, right? Before we dive into the cards, tell me a bit more — what's making you feel stuck? Is it the work itself, the environment, or something else?

Querent: It's mostly that I don't feel challenged anymore. The pay is good but I'm just going through the motions.

Chatbot: Ah, the golden handcuffs situation! That's actually really common. So you're craving growth and meaning, not just a paycheck. Got it. Ready to see what the cards have to say? Focus on your question, take a deep breath, and pick three cards when you're ready.

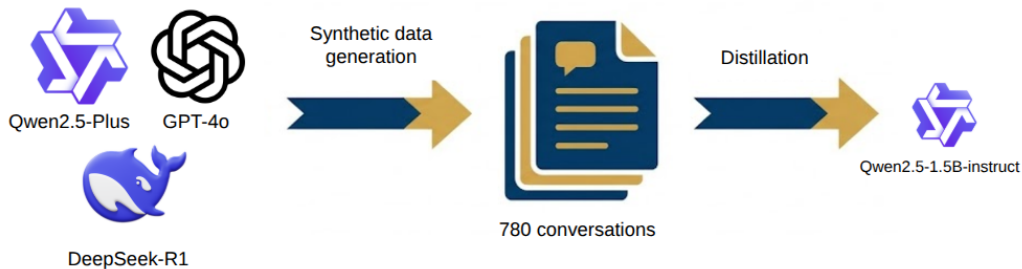
Querent: [Draws: The Tower (reversed), Eight of Cups, The Star]

Chatbot: Ooh, interesting spread! Let's start with The Tower reversed in your first position. Usually The Tower upright is about sudden upheaval, but reversed? It's more like... you're *avoiding* a necessary change. Maybe you already know things need to shake up, but you're holding on because it feels safer?

The Eight of Cups in the middle — this one's literally about walking away from something that no longer fulfills you. See how the figure is leaving those cups behind? That's you, being called to seek something more meaningful...

[continues with card interpretations and synthesis]

Distillation Pipeline: Generating Synthetic Training Data



Data Generation Process:

- ▶ LLMs (Qwen2.5-Plus, GPT-4o, DeepSeek-R1) generate 780 conversations
- ▶ Each conversation includes:
 - ▶ Simulated topics and questions from querents
 - ▶ Specified conversation length
 - ▶ Querent personality traits
 - ▶ Random card combinations with their meanings

Model Selection: Why Qwen2.5-1.5B-Instruct?

Table: Performance on standard benchmarks (Source: Qwen2.5 Technical Report, 2024)

Model	Params	MMLU (Knowledge)	GSM8K (Reasoning)	MATH (Hard Logic)
Llama-3.2-1B-Instruct	1.23B	49.3	44.4	30.6
Gemma-2-2.6B-Base	2.6B	52.2	30.3	25.3
Qwen2.5-0.5B-Instruct	0.49B	47.5	49.6	34.4
Qwen2.5-1.5B-Instruct	1.54B	60.9	73.2	55.2

Reasons for Selection:

- 1. Superior performance-to-parameter ratio
- 2. Strong multilingual capability
- 3. Excellent instruction-following ability
- 4. Efficient inference on consumer hardware

Fine-tuning Results: Comprehensive Evaluation

Table: LLM-as-a-Judge Results (FT = Fine-tuned)

Model	Style	Knowledge	Empathy	Workflow	Coher.	Avg
<i>Judge: Gemma-3-12B-Instruct</i>						
Qwen-0.5B (base)	0.42	0.65	0.48	0.35	0.72	0.52
Qwen-0.5B (FT)	0.78	0.72	0.81	0.85	0.80	0.79
Qwen-1.5B (base)	0.55	0.71	0.58	0.42	0.78	0.61
Qwen-1.5B (FT)	0.88	0.85	0.89	0.92	0.88	0.88
GPT-4 (prompt)	0.82	0.90	0.85	0.75	0.92	0.85
<i>Judge: Mistral-Small-24B-Instruct</i>						
Qwen-1.5B (base)	0.52	0.68	0.55	0.40	0.75	0.58
Qwen-1.5B (FT)	0.85	0.82	0.86	0.90	0.86	0.86
GPT-4o (prompt)	0.80	0.88	0.82	0.72	0.90	0.82

Key Findings:

- ▶ Fine-tuning dramatically improves behavioral metrics, especially workflow compliance (+50%: 0.42 → 0.92)
- ▶ Fine-tuned Qwen-1.5B outperforms prompted GPT-4 on style, empathy, and workflow
- ▶ 100x cost reduction with comparable or better performance

Demo Application using Gradio

Tarot Reading

Instructions

To know about what the cards guide you, you should:

- First, lay in your concerns in the chat.
- Second, chat with the reader so that she can get to know more about your situation.
- Once the reader asks you to draw 3 cards, go down, hit the "Draw and Shuffle Cards" and choose 3 cards, and then return to type "Done" or "Ready" or "Yes", or anything that means you're done, in the chat.
- Then continue to follow the chat to get your full reading.
- Hit the **Reset History** to refresh the chat and ask a new question.

Tap to Chat

Alright, no problem. We can handle everything together then! Before we dive into the reading, though, let's make sure we have three cards drawn. Once you've finished drawing them, type "done" so we can start interpreting them. Sound good?

Done

Perfect! Now, let's begin. The first card we pulled is **Temperance**. This card speaks to balance and harmony—it reminds us that sometimes our biggest challenges arise because we're trying too hard to force change instead of allowing ourselves to adapt naturally. For example, imagine someone who gets stuck in a job because they think they're doing perfectly fine, even though deep down they wish things were different. Does that resonate with you at all? Or perhaps you've experienced similar hesitation before?

What's on your mind? Love, work, family, etc...

Choose 3 cards to reveal your fortune!

When the chatbot tell you to draw cards, hit the shuffle button and draw 3 cards!

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24	C25	C26	C27	C28	C29	C30	C31
C32	C33	C34	C35	C36	C37	C38	C39	C40	C41	C42	C43	C44	C45	C46	C47	C48	C49	C50	C51	C52	C53	C54	C55	C56	C57	C58	C59	C60	C61	C62
C63	C64	C65	C66	C67	C68	C69	C70	C71	C72	C73	C74	C75	C76	C77	C78															



Select Model

old_data_checkpoints/Queen/Queen2.5-1.5B-Instruct-Resumed-33/checkpoint-120

Features:

- ▶ Real-time card selection with visual feedback
- ▶ Multi-turn conversation with context preservation
- ▶ Model switching for comparison

Conclusion and Perspectives

Key Contributions

Problem 1: Advanced RAG Pipeline

- ▶ Developed a structure-preserving multimodal RAG system
- ▶ Achieved **32.8% improvement** in answer correctness on financial documents
- ▶ Achieved **40% improvement** in code hit rate for healthcare documents
- ▶ Demonstrated value of each pipeline component through ablation study

Problem 2: Behavior Adaptation through Fine-tuning

- ▶ Designed an effective distillation pipeline for small models
- ▶ Fine-tuned Qwen-1.5B outperforms prompted GPT-4 on behavioral metrics
- ▶ Achieved **80-90% cost reduction** while maintaining comparable performance
- ▶ Demonstrated feasibility of specialized conversational agents on consumer hardware

Overarching Impact

- ▶ Provided practical solutions to both knowledge and behavior gaps
- ▶ Enabled deployment in resource-constrained, high-stakes environments

Future Directions

Short-term Improvements:

- ▶ Extended evaluation on additional domain-specific datasets
- ▶ Optimization for edge deployment (quantization, pruning)

Long-term Research Directions:

- ▶ **RLHF:** Reinforcement learning from human feedback to further align model outputs with user expectations
- ▶ **Multimodal Generation:** Extending beyond text to support visual explanations (e.g., chart generation in financial reports)
- ▶ **Multilingual Support:** Expanding to support global user bases across multiple languages

Thank You

Q&A Session

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